
Statistics and Statistical Learning

Technical reference

Probability, standard random variables, sampling distributions, estimation, intervals and tests, with proofs. Each chapter says which site note explains it; the last chapter collects the formulas.

Companion to the notes at <https://lorenzoliuzzo.github.io/notes/>

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1 Probability toolkit

In the notes: Probability Foundations · Random Variables, Expectation and Variance

1.1 Probability spaces and the axioms

A probability space is a triple $(\Omega, \mathcal{F}, P)$: a sample space Ω , a σ -algebra $\mathcal{F}$ of events (a family of subsets of Ω containing Ω and closed under complements and countable unions), and a function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the axioms below. Events A, B are *disjoint* if $A \cap B = \emptyset$; events $B_1, B_2, \dots$ form a *partition* if they are pairwise disjoint and $\bigcup_i B_i = \Omega$.

DEFINITION Kolmogorov axioms (A1) $P(A) \geq 0$ for every event A . (A2) $P(\Omega) = 1$. (A3) For pairwise disjoint events $A_1, A_2, \dots$, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

THEOREM Consequences of the axioms For events $A, B, A_1, A_2, \dots$: (i) $P(\emptyset) = 0$; (ii) finite additivity: $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for disjoint A_i ; (iii) $P(A^c) = 1 - P(A)$; (iv) if $A \subseteq B$ then $P(A) \leq P(B)$; in particular $P(A) \leq 1$; (v) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; (vi) union bound: $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (i) Take $A_i = \emptyset$ for all i in (A3): $P(\emptyset) = \sum_i P(\emptyset)$, which forces $P(\emptyset) = 0$. (ii) Apply (A3) with $A_i = \emptyset$ for $i > n$ and use (i). (iii) $\Omega = A \cup A^c$ is a disjoint union, so $1 = P(A) + P(A^c)$. (iv) $B = A \cup (B \setminus A)$ is disjoint, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$; with $B = \Omega$ this gives $P(A) \leq 1$. (v) Write $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$, both disjoint; then $P(B \setminus A) = P(B) - P(A \cap B)$, and substituting gives (v). (vi) Set $B_1 = A_1$ and $B_i = A_i \setminus \bigcup_{j < i} A_j$. The B_i are disjoint, $\bigcup_i B_i = \bigcup_i A_i$ and $B_i \subseteq A_i$, so by (A3) and (iv) $P(\bigcup_i A_i) = \sum_i P(B_i) \leq \sum_i P(A_i)$. $\square$

The general inclusion–exclusion formula is $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots$. Two further facts follow from (A3) by taking limits: if A_n increases to A then $P(A_n) \rightarrow P(A)$, and likewise for decreasing sequences (continuity of P). When Ω is finite and outcomes are equally likely, $P(A) = |A| / |\Omega|$.

1.2 Conditional probability, independence, Bayes

DEFINITION For $P(B) > 0$, the conditional probability of A given B is $P(A | B) = P(A \cap B) / P(B)$. Then $P(\cdot | B)$ is itself a probability: it satisfies (A1)–(A3) on $\mathcal{F}$, so all the consequences above hold conditionally.

Rearranging gives the multiplication rule $P(A \cap B) = P(A | B) P(B)$ and, by induction, the chain rule

$$P(A_1 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) \dots P(A_n | A_1 \cap \dots \cap A_{n-1}). \quad (1)$$

DEFINITION Independence A and B are independent if $P(A \cap B) = P(A) P(B)$ (equivalently $P(A | B) = P(A)$ when $P(B) > 0$). Events $A_1, \dots, A_n$ are *mutually independent* if $P(\bigcap_{i \in S} A_i) = \prod_{i \in S} P(A_i)$ for every subset S of indices; pairwise independence alone does not imply this. A and B are conditionally independent given C if $P(A \cap B | C) = P(A | C) P(B | C)$.

Remark. Disjoint events with positive probability are never independent: $P(A \cap B) = 0 \neq P(A) P(B)$.

THEOREM Law of total probability and Bayes' theorem Let $B_1, B_2, \dots$ be a partition of Ω with $P(B_i) > 0$. For any event A ,

$$P(A) = \sum_i P(A | B_i) P(B_i), \quad (2)$$

and, if $P(A) > 0$,

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_i P(A | B_i) P(B_i)}. \quad (3)$$

Proof. The sets $A \cap B_i$ are disjoint with union A , so by (A3) and the multiplication rule $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A | B_i) P(B_i)$. For Bayes, $P(B_j | A) = P(A \cap B_j) / P(A) = P(A | B_j) P(B_j) / P(A)$, and the denominator is Equation 2. $\square$

In Bayesian language $P(B_j)$ is the prior, $P(A | B_j)$ the likelihood, $P(A)$ the evidence (marginal likelihood) and $P(B_j | A)$ the posterior. The odds form is $P(B | A) / P(B^c | A) = [P(A | B) / P(A | B^c)] \cdot [P(B) / P(B^c)]$: posterior odds equal likelihood ratio times prior odds.

EXAMPLE Diagnostic test Prevalence $P(D) = 0.01$, sensitivity $P(+ | D) = 0.95$, false-positive rate $P(+ | D^c) = 0.10$. Then

$$P(D | +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.088. \quad (4)$$

The posterior is below 9% because the healthy majority produces more false positives than the sick minority produces true ones: the base rate matters.

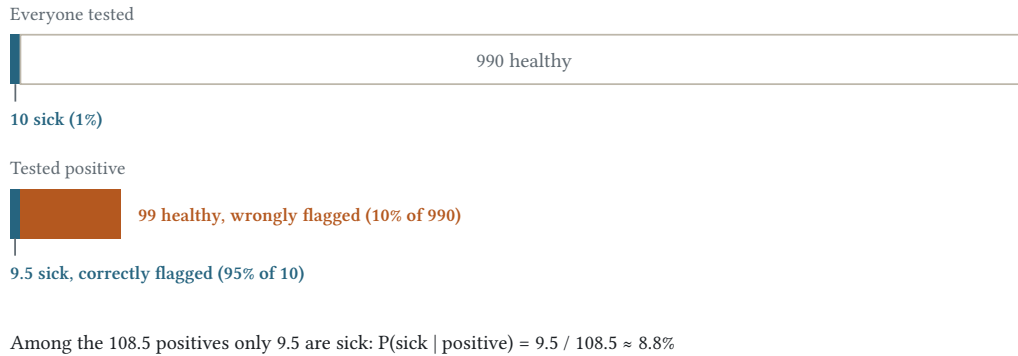


Figure 1: The same test counted in people. The sick are a thin sliver of the population, so the 10% of healthy people who test positive outnumber the true cases about ten to one.

1.3 Random variables and distributions

A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$. Its distribution function is $F(x) = P(X \leq x)$.

LEMMA Properties of the distribution function F is non-decreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. For $a < b$, $P(a < X \leq b) = F(b) - F(a)$.

Proof. Monotonicity is (iv) applied to $\{X \leq x\} \subseteq \{X \leq y\}$. Right-continuity and the limits follow from continuity of P along monotone sequences of events. Finally $\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\}$ is disjoint. $\square$

X is discrete with mass function $p(x) = P(X = x)$, $\sum_x p(x) = 1$, or continuous with density $f = F'$, $\int f = 1$ and $P(a \leq X \leq b) = \int_a^b f$; in the continuous case $P(X = x) = 0$ for every x . The quantile function is $F^{-1}(p) = \inf\{x : F(x) \geq p\}$ and the median is $F^{-1}(1/2)$. For a pair (X, Y) with joint density $f_{X,Y}$, the marginal is $f_{X(x)} = \int f_{X,Y}(x, y) dy$, the conditional density is $f_{X|Y}(x | y) = f_{X,Y}(x, y) / f_Y(y)$, and X, Y are independent iff $f_{X,Y} = f_X f_Y$.

1.4 Expectation

DEFINITION $E[X] = \sum_x xp(x)$ (discrete) or $\int xf(x) dx$ (continuous), when the sum or integral converges absolutely. For a function g (law of the unconscious statistician), $E[g(X)] = \sum_x g(x)p(x)$ or $\int g(x)f(x) dx$.

THEOREM Properties of expectation (i) Linearity: $E[aX + bY + c] = aE[X] + bE[Y] + c$ for any X, Y , dependent or not. (ii) Monotonicity: $X \leq Y$ implies $E[X] \leq E[Y]$. (iii) If X, Y are independent, $E[XY] = E[X]E[Y]$. (iv) $E[\mathbb{1}_A] = P(A)$. (v) Jensen: for convex g , $g(E[X]) \leq E[g(X)]$.

Proof. (i) and (ii) follow from linearity and monotonicity of sums and integrals, using the joint density for X and Y together. (iii) By independence $f_{X,Y} = f_X f_Y$, so $E[XY] = \int \int xy f_X(x) f_Y(y) dx dy = E[X]E[Y]$. (iv) $\mathbb{1}_A$ takes the value 1 with probability $P(A)$ and 0 otherwise. (v) A convex g lies above its tangent line at $m = E[X]$: $g(x) \geq g(m) + c(x - m)$ for some slope c ; take expectations. $\square$

1.5 Variance, covariance, correlation

DEFINITION $\text{Var}(X) = E[(X - \mu)^2]$ with $\mu = E[X]$; the standard deviation is $\sigma = \sqrt{\text{Var}(X)}$. $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$ and, when $\sigma_X, \sigma_Y > 0$, the correlation is $\rho_{XY} = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$. Moments: $E[X^k]$; central moments $E[(X - \mu)^k]$; skewness $E[(X - \mu)^3]/\sigma^3$ and kurtosis $E[(X - \mu)^4]/\sigma^4$ (equal to 3 for the normal).

THEOREM Properties of variance and covariance (i) $\text{Var}(X) = E[X^2] - (E[X])^2 \geq 0$, with equality iff X is almost surely constant. (ii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$. (iii) $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$, $\text{Cov}(X, X) = \text{Var}(X)$, and Cov is bilinear. (iv) $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$. (v) Independent X, Y have $\text{Cov}(X, Y) = 0$; the converse is false. (vi) $|\rho_{XY}| \leq 1$, with equality iff $Y = aX + b$ almost surely.

Proof. (i) Expand $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$ and take expectations; nonnegativity holds because the integrand is ≥ 0 , and $E[(X - \mu)^2] = 0$ forces $X = \mu$ almost surely. (ii) $aX + b - E[aX + b] = a(X - \mu)$. (iii) Expand the product and use linearity; bilinearity is linearity in each argument. (iv) is (ii) and (iii) combined. (v) follows from (iii) and property (iii) of expectation. For the failure of the converse, let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$: $\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$ since odd moments of a symmetric law vanish, yet Y is a function of X . (vi) For every real t , $0 \leq \text{Var}(Y - tX) = \text{Var}(Y) - 2t \text{Cov}(X, Y) + t^2 \text{Var}(X)$. A quadratic in t that never goes negative has non-positive discriminant, $4\text{Cov}(X, Y)^2 - 4 \text{Var}(X) \text{Var}(Y) \leq 0$, which is $|\rho| \leq 1$. Equality means the discriminant vanishes, so $\text{Var}(Y - tX) = 0$ for some t and $Y - tX$ is constant. $\square$

Property (iv) generalizes to $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$, and so, for uncorrelated (hence independent) X_i with common variance σ^2 ,

$$\text{Var}\left(\sum_i X_i\right) = n\sigma^2, \quad \text{Var}(\bar{X}_n) = \sigma^2/n. \quad (5)$$

THEOREM Law of total expectation and total variance If $E[X^2] < \infty$, then $E[X] = E[E[X | Y]]$ and $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$. $\square$

Proof. The first identity is the law of total probability for expectations: $E[X] = \int E[X | Y = y] f_Y(y) dy$. For the second, write $X - E[X] = (X - E[X | Y]) + (E[X | Y] - E[X])$ and square. The cross term has expectation zero: conditionally on Y the second factor is fixed and $E[X - E[X | Y] | Y] = 0$. The first square has expectation $E[\text{Var}(X | Y)]$ and the second $\text{Var}(E[X | Y])$. $\square$

1.6 Markov and Chebyshev

LEMMA Markov and Chebyshev If $X \geq 0$ and $a > 0$ then $P(X \geq a) \leq E[X]/a$. For any X with mean μ and variance σ^2 and any $k > 0$, $P(|X - \mu| \geq k) \leq \sigma^2/k^2$.

Proof. $a \mathbb{1}\{X \geq a\} \leq X$ pointwise, take expectations. Chebyshev is Markov applied to $(X - \mu)^2$ and $a = k^2$. $\square$

The standard discrete and continuous distributions, with their means, variances and uses, are in the chapters on discrete and continuous random variables below.

1.7 Limit theorems

Write $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

THEOREM Weak law of large numbers If X_i are i.i.d. with $E X_1 = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$, then $\bar{X}_n \xrightarrow{P} \mu$, i.e. $P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Proof. By Equation 5, $\text{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev gives $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$. $\square$

THEOREM Central limit theorem If X_i are i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$, then

$$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1). \quad (7)$$

The proof goes through characteristic functions and is not reproduced here. The Berry–Esseen theorem quantifies the rate: the Kolmogorov distance to Φ is at most $C E|X_1 - \mu|^3 / (\sigma^3 \sqrt{n})$. This is why the sample size needed depends on skewness.

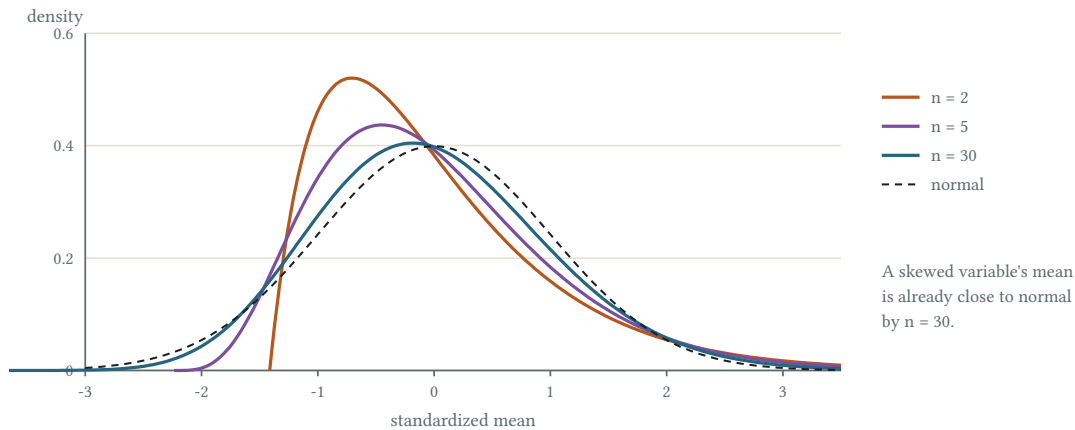


Figure 2: Standardized mean of n exponential variables against the standard normal density. At $n = 2$ the skew is obvious, by $n = 30$ the two curves are hard to tell apart.

RULE OF THUMB $n \geq 30$ is a guide, not a theorem: the Berry–Esseen bound grows with skewness, and for heavy-tailed data it takes far more.

THEOREM Slutsky and delta method If $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{P} b$ (constant), then $A_n + B_n \xrightarrow{d} A + b$ and $A_n B_n \xrightarrow{d} bA$. If moreover $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \tau^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then

$$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, g'(\theta)^2 \tau^2). \quad (8)$$

Slutsky is what licenses replacing an unknown σ by a consistent estimate in a pivot, which is how the Wald and large-sample t intervals are justified.

2 Discrete random variables

In the notes: Common Random Variables

A discrete random variable has mass function $p(x) = P(X = x)$ on a countable support, with $\sum_x p(x) = 1$, $E[X] = \sum_x xp(x)$ and $\text{Var}(X) = E[X^2] - (E[X])^2$. Probability background is in the Probability toolkit chapter.

2.1 Bernoulli and binomial

DEFINITION $X \sim \text{Bernoulli}(p)$ if $P(X = 1) = p$ and $P(X = 0) = 1 - p$. $Y \sim \text{Binomial}(n, p)$ is the sum of n independent $\text{Bernoulli}(p)$ variables, with mass function $P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $k = 0, \dots, n$.

THEOREM Mean and variance If $X \sim \text{Bernoulli}(p)$ then $E[X] = p$ and $\text{Var}(X) = p(1 - p)$. If $Y \sim \text{Binomial}(n, p)$ then $E[Y] = np$ and $\text{Var}(Y) = np(1 - p)$.

Proof. $E[X] = 1 \cdot p + 0 \cdot (1 - p) = p$, and since $X^2 = X$, $E[X^2] = p$ and $\text{Var}(X) = p - p^2$. For $Y = \sum_{i=1}^n X_i$ with independent X_i , linearity gives $E[Y] = np$ and independence gives $\text{Var}(Y) = \sum_i \text{Var}(X_i) = np(1 - p)$. $\square$

The binomial mass function counts the $\binom{n}{k}$ arrangements of k successes among n trials, each with probability $p^k(1 - p)^{n-k}$. For large n it is approximately $\mathcal{N}(np, np(1 - p))$ by the central limit theorem; a common rule is $np(1 - p) \geq 10$.

2.2 Geometric

DEFINITION $X \sim \text{Geometric}(p)$ counts the trials up to and including the first success: $P(X = k) = (1 - p)^{k-1} p$, $k = 1, 2, \dots$.

THEOREM Mean and memorylessness $E[X] = 1/p$ and $\text{Var}(X) = (1 - p)/p^2$. Moreover $P(X > m + k \mid X > m) = P(X > k)$ for all integers $m, k \geq 0$.

Proof. With $q = 1 - p$, $E[X] = p \sum_{k \geq 1} k q^{k-1} = p/(1 - q)^2 = 1/p$, using $\sum_{k \geq 1} k q^{k-1} = \frac{d}{dq} \sum_{k \geq 0} q^k = (1 - q)^{-2}$. A second differentiation gives $E[X(X - 1)]$ and hence the variance. For the tail, $P(X > k) = q^k$, so $P(X > m + k \mid X > m) = q^{m+k}/q^m = q^k$. $\square$

2.3 Poisson

DEFINITION $X \sim \text{Poisson}(\lambda)$ has $P(X = k) = e^{-\lambda} \lambda^k / k!$, $k = 0, 1, 2, \dots$.

THEOREM Moments, sums and the binomial limit (i) $E[X] = \text{Var}(X) = \lambda$. (ii) If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent then $X + Y \sim \text{Poisson}(\lambda + \mu)$. (iii) If $X_n \sim \text{Binomial}(n, \lambda/n)$ then $P(X_n = k) \rightarrow e^{-\lambda} \lambda^k / k!$ for every fixed k .

Proof. (i) $E[X] = \sum_{k \geq 1} k e^{-\lambda} \lambda^k / k! = \lambda e^{-\lambda} \sum_{k \geq 1} \lambda^{k-1} / (k-1)! = \lambda$, and similarly $E[X(X - 1)] = \lambda^2$, so $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$. (ii) $P(X + Y = m) = \sum_{j=0}^m e^{-\lambda} \lambda^j / j! \cdot e^{-\mu} \mu^{m-j} / (m - j)!$

$j)! = e^{-(\lambda+\mu)}(\lambda+\mu)^m/m!$ by the binomial theorem. (iii) $P(X_n = k) = \frac{n!}{(n-k)!k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$. As $n \rightarrow \infty$ the first factor tends to 1, $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$ and $(1 - \lambda/n)^{-k} \rightarrow 1$. $\square$

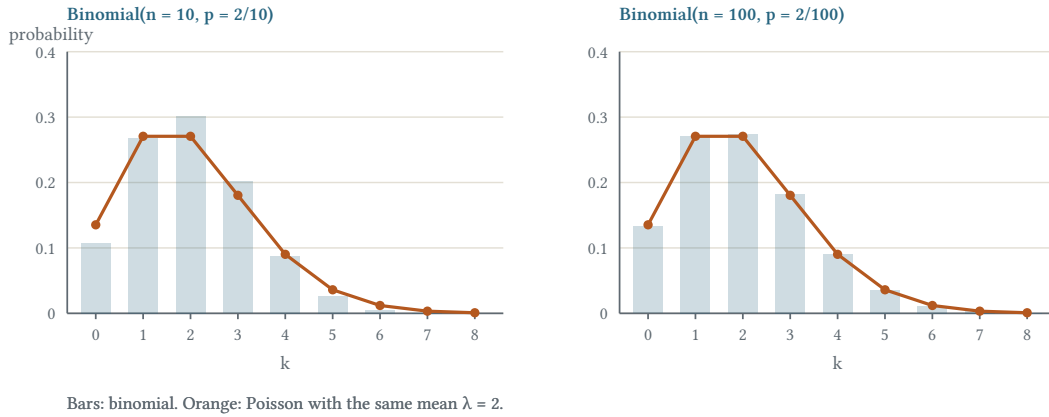


Figure 3: Binomial bars with the Poisson of the same mean $\lambda = 2$ on top. With $n = 10$ they differ visibly, with $n = 100$ they nearly coincide.

3 Continuous random variables

In the notes: Common Random Variables

A continuous random variable has a density f with $P(a \leq X \leq b) = \int_a^b f$, mean $E[X] = \int x f(x) dx$ and distribution function $F(x) = \int_{-\infty}^x f$.

3.1 Uniform and inverse transform sampling

$X \sim \text{Uniform}(a, b)$ has $f(x) = 1/(b-a)$ on $[a, b]$, $E[X] = (a+b)/2$ and $\text{Var}(X) = (b-a)^2/12$ (compute $E[X^2] = (a^2 + ab + b^2)/3$).

THEOREM Inverse transform Let F be a distribution function, $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, and $U \sim \text{Uniform}(0, 1)$. Then $F^{-1}(U)$ has distribution function F . Conversely, if X has a continuous distribution function F , then $F(X) \sim \text{Uniform}(0, 1)$.

Proof. The generalized inverse satisfies $F^{-1}(u) \leq x$ iff $u \leq F(x)$, so $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$. For the converse, with F continuous and strictly increasing on the support, $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$. $\square$

The converse is the probability integral transform used for the uniformity of p -values; the direct statement is how simulation software turns uniform random numbers into draws from any distribution with a computable F^{-1} .

3.2 Exponential

DEFINITION $X \sim \text{Exponential}(\lambda)$ has $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$, so that $P(X > x) = e^{-\lambda x}$.

THEOREM Moments and memorylessness $E[X] = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$, and $P(X > s+t \mid X > s) = P(X > t)$.

Proof. Integrating by parts, $E[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = 1/\lambda$ and $E[X^2] = 2/\lambda^2$, so $\text{Var}(X) = 1/\lambda^2$. For the tail, $P(X > s+t \mid X > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. $\square$

The exponential is the only continuous distribution on $[0, \infty)$ with this property. If events arrive as a Poisson process of rate λ (the count in an interval of length t is $\text{Poisson}(\lambda t)$), the waiting time to the first event, and between successive events, is $\text{Exponential}(\lambda)$: $P(T > t) = P(\text{no event in } [0, t]) = e^{-\lambda t}$.

3.3 Normal

DEFINITION $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x - \mu)^2/(2\sigma^2))$. The standard normal $\mathcal{N}(0, 1)$ has distribution function Φ and quantiles $z_p = \Phi^{-1}(p)$.

THEOREM Standardization and closure (i) $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$. (ii) $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$, so $P(X \leq x) = \Phi((x - \mu)/\sigma)$. (iii) If $X_1, \dots, X_n$ are independent, $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $\sum_i a_i X_i + b \sim \mathcal{N}(\sum_i a_i \mu_i + b, \sum_i a_i^2 \sigma_i^2)$.

Proof. (ii) The substitution $y = (u - \mu)/\sigma$ in $\int_{-\infty}^x f(u) du$ gives $\int_{-\infty}^{(x-\mu)/\sigma} \varphi(y) dy$. (i) follows from (ii) since $E[Z] = 0$ (odd integrand) and $E[Z^2] = 1$ (integration by parts). (iii) The mean and variance follow from linearity and independence; that the sum is again normal follows from the moment generating function $E[e^{tX}] = e^{\mu t + \sigma^2 t^2/2}$, which multiplies for independent summands. $\square$

Quantiles that recur: $z_{0.90} \approx 1.282$, $z_{0.95} \approx 1.645$, $z_{0.975} \approx 1.960$, $z_{0.995} \approx 2.576$, and $P(|Z| \leq 1, 2, 3) \approx 0.683, 0.954, 0.997$.

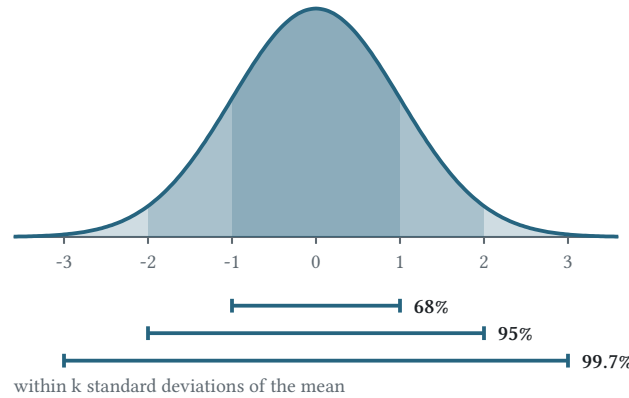


Figure 4: The 68, 95 and 99.7 percent rule.

3.4 Gamma and chi-square

DEFINITION $X \sim \text{Gamma}(\alpha, \beta)$ (shape $\alpha > 0$, rate $\beta > 0$) has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ on $x > 0$, with $E[X] = \alpha/\beta$ and $\text{Var}(X) = \alpha/\beta^2$. The sum of k independent $\text{Exponential}(\lambda)$ variables is $\text{Gamma}(k, \lambda)$. The chi-square distribution is $\chi_k^2 = \text{Gamma}(k/2, 1/2)$, with density $x^{k/2-1} e^{-x/2} / (2^{k/2} \Gamma(k/2))$.

THEOREM Chi-square as a sum of squares If $Z_1, \dots, Z_k$ are independent $\mathcal{N}(0, 1)$ then $\sum_i Z_i^2 \sim \chi_k^2$, with mean k and variance $2k$.

Proof. The density of Z_1^2 follows by the change of variable $y = z^2$ and equals the χ_1^2 density; independent Gamma variables with a common rate add their shapes, giving χ_k^2 . For the moments, $E[Z^2] = 1$ and $\text{Var}(Z^2) = E[Z^4] - 1 = 3 - 1 = 2$, and the Z_i^2 are independent, so the mean is k and the variance $2k$. $\square$

4 Sampling distributions of statistics

In the notes: Sampling Distributions

A statistic $T = T(X_1, \dots, X_n)$ is a random variable. Its sampling distribution is the law of T when the sample is drawn from the assumed model, and its standard error is $\text{se}(T) = \sqrt{\text{Var}(T)}$.

4.1 Mean, variance and proportion

THEOREM Mean and variance of $\bar{X}$ For i.i.d. X_i with mean μ and variance σ^2 , $\mathbb{E}[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$. For two independent samples, $\bar{X}_1 - \bar{X}_2$ has mean $\mu_1 - \mu_2$ and variance $\sigma_1^2/n_1 + \sigma_2^2/n_2$.

Proof. Linearity gives $\mathbb{E}[\bar{X}] = n^{-1} \sum_i \mathbb{E}[X_i] = \mu$; independence gives $\text{Var}(\bar{X}) = n^{-2} \sum_i \text{Var}(X_i) = \sigma^2/n$. For the difference, $\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)$ by independence. $\square$

If the data are normal, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly; in general the central limit theorem gives $\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$.

THEOREM Sample proportion If $Y \sim \text{Binomial}(n, p)$ and $\hat{p} = Y/n$, then $\mathbb{E}[\hat{p}] = p$, $\text{Var}(\hat{p}) = p(1-p)/n$, and $\sqrt{n}(\hat{p} - p)/\sqrt{p(1-p)} \xrightarrow{d} \mathcal{N}(0, 1)$.

Proof. The moments follow from those of the binomial and the scaling rule $\text{Var}(Y/n) = \text{Var}(Y)/n^2$. The limit is the central limit theorem applied to $Y = \sum_i X_i$ with Bernoulli summands (the de Moivre–Laplace theorem). $\square$

EXAMPLE Classifier accuracy A classifier with true accuracy $p = 0.8$ evaluated on $n = 100$ independent test examples has $\text{se}(\hat{p}) = \sqrt{0.8 \cdot 0.2/100} = 0.04$, so the observed accuracy lies in 0.8 ± 0.08 with probability about 95%.

4.2 Normal-theory distributions

DEFINITION Gaussian family $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x - \mu)^2/(2\sigma^2))$. Let $Z_1, \dots, Z_k \sim \mathcal{N}(0, 1)$ be independent. Then

- $\sum_i Z_i^2 \sim \chi_k^2$ (chi-square, k degrees of freedom), with mean k and variance $2k$;
- if $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_k^2$ are independent, $Z/\sqrt{V/k} \sim t_k$ (Student);
- if $V_1 \sim \chi_a^2$ and $V_2 \sim \chi_b^2$ are independent, $(V_1/a)/(V_2/b) \sim F_{a,b}$.

Facts used below: a linear combination of jointly Gaussian variables is Gaussian; for jointly Gaussian variables, uncorrelated implies independent; $t_k \rightarrow \mathcal{N}(0, 1)$ as $k \rightarrow \infty$; $t_k^2 \sim F_{1,k}$.

Let $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d., with $\bar{X} = \bar{X}_n$ and $S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2$.

THEOREM Distribution of $\bar{X}$ and S^2 (i) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$. (ii) $\bar{X}$ and S^2 are independent. (iii) $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. (iv) Consequently

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}. \quad (9)$$

Proof. (i) is closure of the Gaussian family under linear maps. For (ii)–(iii), let $Z_i = (X_i - \mu)/\sigma$ and Q an orthogonal $n \times n$ matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$. Then $Y = QZ$ has $Y_1 = \sqrt{n}\bar{Z}$ and, since Q

is orthogonal, $Y \sim \mathcal{N}(0, I_n)$: the Y_i are independent standard normals. Orthogonality preserves the norm: $\sum_i Z_i^2 = \sum_i Y_i^2$, and $\sum_i Z_i^2 = \sum_i (Z_i - \bar{Z})^2 + n\bar{Z}^2$, hence

$$\sum_i (Z_i - \bar{Z})^2 = \sum_{i=2}^n Y_i^2 \sim \chi_{n-1}^2, \quad (10)$$

a function of $(Y_2, \dots, Y_n)$ only, and so independent of Y_1 and thus of $\bar{Z}$. Since $\sum_i (Z_i - \bar{Z})^2 = (n-1)S^2/\sigma^2$ and $\bar{X} = \mu + \sigma\bar{Z}$, (ii) and (iii) follow. For (iv), write $T = [(\bar{X} - \mu)/(\sigma/\sqrt{n})] / \sqrt{((n-1)S^2/\sigma^2)/(n-1)}$, a standard normal over the root of an independent $\chi_{n-1}^2/(n-1)$. $\square$

THEOREM Unbiasedness of S^2 For any i.i.d. sample with finite variance σ^2 , $E[S^2] = \sigma^2$.

Proof. $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations, $E[\sum_i (X_i - \bar{X})^2] = n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$. $\square$

4.3 The t and F distributions in more detail

The t_k distribution has density $\Gamma((k+1)/2)/(\sqrt{k\pi}\Gamma(k/2)) \cdot (1+t^2/k)^{-(k+1)/2}$. It is symmetric about 0, has mean 0 for $k > 1$ and variance $k/(k-2)$ for $k > 2$, and has heavier tails than $\mathcal{N}(0, 1)$, to which it converges as $k \rightarrow \infty$. Selected 97.5% quantiles are $t_{5,0.975} = 2.571$, $t_{9,0.975} = 2.262$, $t_{29,0.975} = 2.045$, against $z_{0.975} = 1.960$.

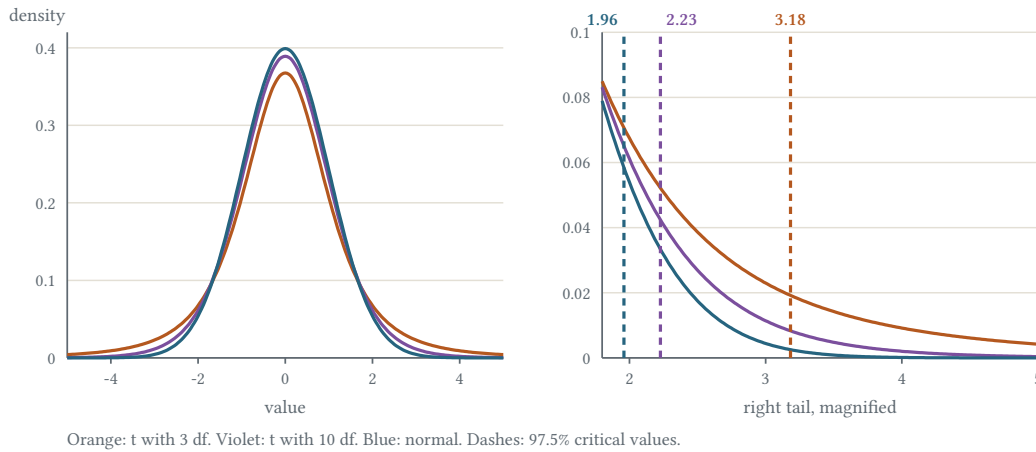


Figure 5: Left, the whole density: the normal and two t curves look alike. Right, the right tail magnified: the t puts more mass out there, so its critical value is larger and its intervals are wider.

LEMMA Relation between t and F If $T \sim t_k$ then $T^2 \sim F_{1,k}$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim \mathcal{N}(0, 1)$ independent of $V \sim \chi_k^2$. Then $T^2 = Z^2/(V/k) = (Z^2/1)/(V/k)$ with $Z^2 \sim \chi_1^2$, the ratio defining $F_{1,k}$. $\square$

The $F_{a,b}$ distribution has mean $b/(b-2)$ for $b > 2$ and is the null distribution of the ratio of two independent normal-sample variance estimates, S_1^2/S_2^2 scaled by σ_2^2/σ_1^2 , with $a = n_1 - 1$ and $b = n_2 - 1$; it is also the null distribution of the overall test in analysis of variance and linear regression.

4.4 A non-normal example: the sample maximum

THEOREM Maximum of uniform samples Let $X_1, \dots, X_n$ be i.i.d. $\text{Uniform}(0, \theta)$ and $M = \max_i X_i$. Then $P(M \leq m) = (m/\theta)^n$ for $0 \leq m \leq \theta$, with density nm^{n-1}/θ^n , $E[M] = n\theta/(n+1)$, and $(n+1)M/n$ is unbiased for θ .

Proof. $M \leq m$ iff every $X_i \leq m$, so by independence $P(M \leq m) = (m/\theta)^n$. Differentiating gives the density, and $E[M] = \int_0^\theta m \cdot nm^{n-1}/\theta^n dm = n\theta/(n+1)$. $\square$

The sampling distribution of M is concentrated against the boundary θ and is far from normal, so normal-theory intervals do not apply: sampling distributions must be derived for the statistic actually used.

4.5 Simulation and the bootstrap

Any sampling distribution can be approximated by simulation: draw B samples from the assumed model, compute T on each, and use the empirical distribution of $T_1, \dots, T_B$; by the law of large numbers its mean and standard deviation converge to $E[T]$ and $se(T)$ as $B \rightarrow \infty$.

When the model is unknown, the bootstrap replaces the unknown distribution F by the empirical distribution $\hat{F}_n$, which puts mass $1/n$ on each observation. Resample n observations with replacement, compute T_b^* , repeat for $b = 1, \dots, B$, and estimate $se(T)$ by the standard deviation of the T_b^* . It is consistent for smooth statistics such as the mean; it fails for extreme statistics such as the maximum in the previous example.

5 Point estimation

In the notes: Point Estimation and Maximum Likelihood

This part uses the probability toolkit (axioms, expectation and variance rules, limit theorems, Slutsky) and the sampling distributions of the mean and variance, the χ^2 , t and F laws, from the chapters above. They are quoted without proof.

Let $X_1, \dots, X_n \sim f(\cdot; \theta)$, $\theta \in \Theta \subset \mathbb{R}$, and $\hat{\theta} = T(X_1, \dots, X_n)$.

DEFINITION $\text{bias}(\hat{\theta}) = E_\theta[\hat{\theta}] - \theta$. The estimator is unbiased if this vanishes for all θ . It is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$. Its mean squared error is $\text{MSE}(\hat{\theta}) = E_\theta[(\hat{\theta} - \theta)^2]$.

THEOREM Bias–variance decomposition

$$\text{MSE}(\hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{bias}(\hat{\theta})^2. \quad (11)$$

Proof. Add and subtract $E\hat{\theta}$: $(\hat{\theta} - \theta)^2 = (\hat{\theta} - E\hat{\theta})^2 + 2(\hat{\theta} - E\hat{\theta})(E\hat{\theta} - \theta) + (E\hat{\theta} - \theta)^2$. The cross term has zero expectation because $E\hat{\theta} - \theta$ is a constant and $E[\hat{\theta} - E\hat{\theta}] = 0$. $\square$

EXAMPLE A biased estimator with smaller MSE For $X_i \sim \mathcal{N}(\mu, \sigma^2)$, the estimator $\hat{\sigma}_c^2 = c \sum_i (X_i - \bar{X})^2$ has, using $\sum_i (X_i - \bar{X})^2 \sim \sigma^2 \chi_{n-1}^2$ (mean $(n-1)\sigma^2$, variance $2(n-1)\sigma^4$),

$$\text{MSE} = \sigma^4[2c^2(n-1) + ((n-1)c - 1)^2]. \quad (12)$$

This is minimized at $c = 1/(n+1)$, not at the unbiased choice $c = 1/(n-1)$. The unbiased estimator is therefore not MSE-optimal, a first instance of the shrinkage idea behind Ridge and Lasso.

5.1 Fisher information and the Cramér–Rao bound

Assume the regularity conditions: the support of $f(\cdot; \theta)$ does not depend on θ , $\log f$ is twice differentiable in θ , and differentiation under the integral sign is allowed.

DEFINITION Score and Fisher information The score of one observation is $s(x; \theta) = \partial_\theta \log f(x; \theta)$, and

$$I(\theta) = \mathbb{E}_\theta[s(X; \theta)^2] = \text{Var}_\theta(s(X; \theta)). \quad (13)$$

LEMMA Under the regularity conditions, $\mathbb{E}_\theta[s(X; \theta)] = 0$ and $I(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \log f(X; \theta)]$. For n independent observations the information is $nI(\theta)$.

Proof. $\mathbb{E}[s] = \int (\partial_\theta f / f) f \, dx = \partial_\theta \int f \, dx = \partial_\theta 1 = 0$. Differentiating $\int s f \, dx = 0$ once more gives $\int (\partial_\theta s) f \, dx + \int s^2 f \, dx = 0$, and $\partial_\theta s = \partial_\theta^2 \log f$. Additivity follows because the log-likelihood of independent data is the sum of the individual ones, so the scores add and, being independent with mean zero, their variances add. $\square$

THEOREM Cramér–Rao lower bound If $\hat{\theta}$ is unbiased with finite variance, then $\text{Var}_\theta(\hat{\theta}) \geq 1/(nI(\theta))$.

Proof. Let $U = \sum_i s(X_i; \theta)$ be the total score, so $\mathbb{E} U = 0$ and $\text{Var}(U) = nI(\theta)$. Unbiasedness reads $\int \hat{\theta}(x) f_{n(x; \theta)} \, dx = \theta$ (f_n the joint density). Differentiating in θ gives $\int \hat{\theta}(x) U(x) f_n \, dx = 1$, i.e. $\text{Cov}(\hat{\theta}, U) = \mathbb{E}[\hat{\theta} U] = 1$ since $\mathbb{E} U = 0$. Cauchy–Schwarz: $1 = \text{Cov}(\hat{\theta}, U)^2 \leq \text{Var}(\hat{\theta}) \text{Var}(U) = \text{Var}(\hat{\theta}) nI(\theta)$. $\square$

Remark. For a biased estimator with $b(\theta) = \text{bias}(\hat{\theta})$ the same argument gives $\text{Var}(\hat{\theta}) \geq (1 + b'(\theta))^2 / (nI(\theta))$. Equality in Cauchy–Schwarz holds iff $\hat{\theta} - \theta$ is proportional to the score, which happens exactly in exponential families for the mean-value parameter.

5.2 Sufficiency (brief)

A statistic $T(X)$ is sufficient for θ if the conditional law of the sample given T does not depend on θ . By the Fisher–Neyman factorization theorem, T is sufficient iff $f_{n(x; \theta)} = g(T(x); \theta)h(x)$. For normal data with known σ , $\sum_i X_i$ is sufficient for μ ; for the Bernoulli model, $\sum_i X_i$ is sufficient for p . The Rao–Blackwell theorem states that conditioning an estimator on a sufficient statistic never increases its MSE.

6 Maximum likelihood

In the notes: Point Estimation and Maximum Likelihood

DEFINITION The likelihood is $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$, the log-likelihood $\ell(\theta) = \log L(\theta)$ and the maximum likelihood estimator $\hat{\theta}_{\text{MLE}} = \arg \max_\theta \ell(\theta)$.

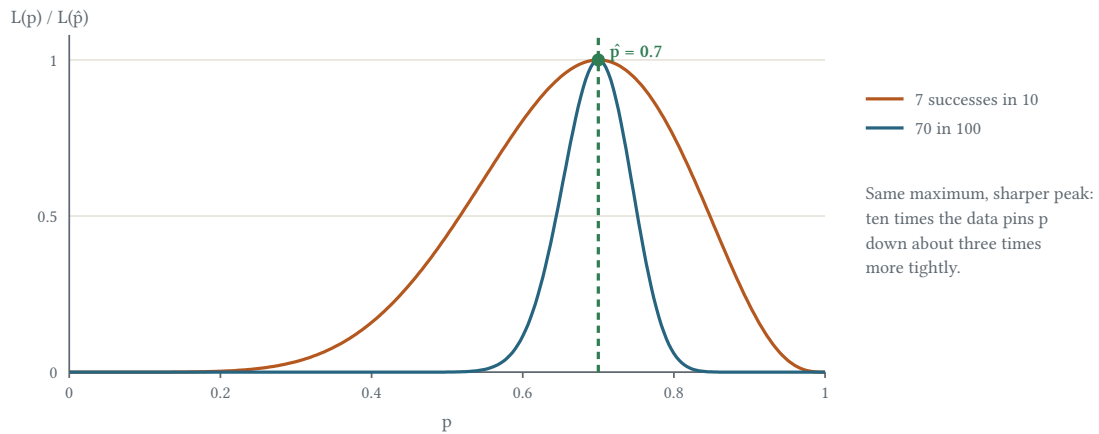


Figure 6: Likelihood of a success probability for the same observed proportion at two sample sizes, each scaled to peak at 1. The maximizer is the same; the curvature at the peak, which is the Fisher information, grows with n .

EXAMPLE Bernoulli $\ell(p) = (\sum x_i) \log p + (n - \sum x_i) \log(1 - p)$, $\ell'(p) = (\sum x_i)/p - (n - \sum x_i)/(1 - p)$. Setting it to zero gives $\hat{p} = \bar{x}$; $\ell'' < 0$ so it is a maximum. The information is $I(p) = 1/(p(1 - p))$ and $\text{Var}(\hat{p}) = p(1 - p)/n = 1/(nI(p))$: the MLE attains the Cramér–Rao bound exactly.

EXAMPLE Normal $\ell(\mu, \sigma^2) = -(n/2) \log(2\pi\sigma^2) - (2\sigma^2)^{-1} \sum_i (x_i - \mu)^2$. The first-order conditions give $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = n^{-1} \sum_i (x_i - \bar{x})^2$. By the theorem on S^2 , $E \hat{\sigma}^2 = ((n - 1)/n)\sigma^2$: the MLE of the variance is biased, but consistent, and $\hat{\sigma}^2 = cS^2$ with $c = (n - 1)/n \rightarrow 1$.

6.1 Asymptotic properties

THEOREM Asymptotics of the MLE Under the regularity conditions above, with θ_0 the true parameter and $I(\theta_0) > 0$: (i) $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta_0$; (ii) $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$; (iii) invariance: the MLE of $g(\theta)$ is $g(\hat{\theta}_{\text{MLE}})$, and by the delta method it is asymptotically $\mathcal{N}(g(\theta_0), g'(\theta_0)^2/(nI(\theta_0)))$.

Proof. Sketch of (ii). Expand the score at the MLE around θ_0 : $0 = \ell'(\hat{\theta}) = \ell'(\theta_0) + (\hat{\theta} - \theta_0)\ell''(\tilde{\theta})$ for some $\tilde{\theta}$ between them. Hence

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{n^{-1/2}\ell'(\theta_0)}{-n^{-1}\ell''(\tilde{\theta})}. \quad (14)$$

The numerator is $n^{-1/2} \sum_i s(X_i; \theta_0)$, a normalized sum of i.i.d. mean-zero variables with variance $I(\theta_0)$, so by the CLT it tends to $\mathcal{N}(0, I(\theta_0))$. By the law of large numbers and consistency, the denominator tends to $-E[\partial_\theta^2 \log f] = I(\theta_0)$. Slutsky gives the limit $\mathcal{N}(0, I(\theta_0)^{-1})$. $\square$

The practical consequence is the standard error $\hat{\text{se}} = 1/\sqrt{nI(\hat{\theta})}$ (or its observed-information version, $1/\sqrt{-\ell''(\hat{\theta})}$), valid for large n .

6.2 Method of moments

Equate the first d population moments $m_{j(\theta)} = E_\theta X^j$ to the sample moments $n^{-1} \sum_i X_i^j$ and solve for $\theta \in \mathbb{R}^d$. The estimator is consistent when $m = (m_1, \dots, m_d)$ is invertible with continuous inverse, but generally has larger asymptotic variance than the MLE.

7 Interval estimation

In the notes: Confidence Intervals

DEFINITION A $(1 - \alpha)$ confidence interval is a random interval $[L(X), U(X)]$ with $P_\theta(L(X) \leq \theta \leq U(X)) \geq 1 - \alpha$ for all $\theta \in \Theta$. The left side is the coverage; the inequality is replaced by an equality for exact intervals.

The probability refers to the random endpoints under repeated sampling. For the realized interval, θ is fixed and the event either holds or not.

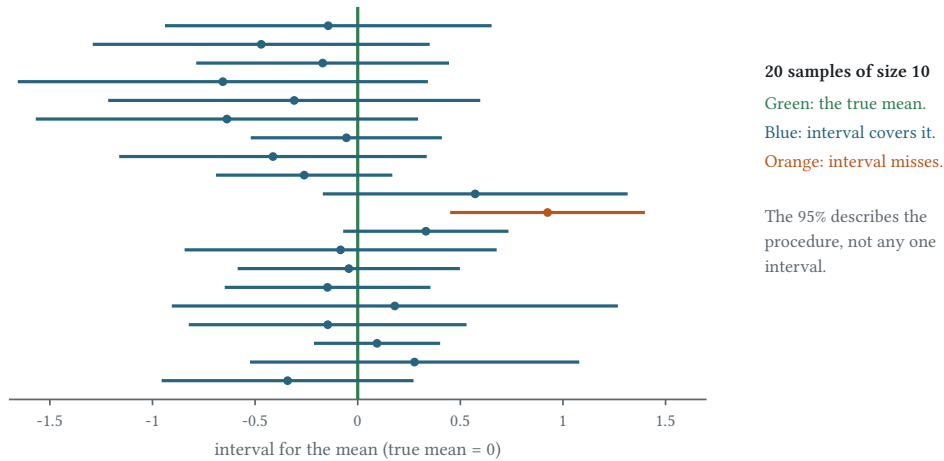


Figure 7: Twenty independent samples, each with its own 95% interval for the same true mean. The intervals vary, the truth does not; about one in twenty misses, here the orange one.

WATCH OUT The 95% describes the procedure, not the one interval you computed: it is not the probability that θ lies inside it.

DEFINITION Pivot $Q(X; \theta)$ is a pivot if its distribution does not depend on θ .

If $P(a \leq Q(X; \theta) \leq b) = 1 - \alpha$ and Q is monotone in θ , solving $a \leq Q \leq b$ for θ gives an exact interval.

THEOREM Intervals for the normal model Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d.. Exact $(1 - \alpha)$ intervals:

- mean, σ unknown: $\bar{X} \pm t_{n-1, 1-\alpha/2} S / \sqrt{n}$;
- mean, σ known: $\bar{X} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$;
 $[(n-1)S^2 / \chi_{n-1, 1-\alpha/2}^2, (n-1)S^2 / \chi_{n-1, \alpha/2}^2]$;
- variance: $(n-1)S^2 / \chi_{n-1, \alpha/2}^2$.

Proof. $P(-t \leq T \leq t) = 1 - \alpha$ with $t = t_{n-1, 1-\alpha/2}$ and T the pivot of the sampling-distribution theorem; rearranging $-t \leq (\bar{X} - \mu) / (S / \sqrt{n}) \leq t$ gives the first interval. The second is identical with T replaced by $\sqrt{n}(\bar{X} - \mu) / \sigma \sim \mathcal{N}(0, 1)$. For the third, $P(\chi_{n-1, \alpha/2}^2 \leq (n-1)S^2 / \sigma^2 \leq \chi_{n-1, 1-\alpha/2}^2) = 1 - \alpha$ and inverting in σ^2 reverses the inequalities. $\square$

THEOREM Wald interval If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta))$ with v continuous, then $\hat{\theta} \pm z_{1-\alpha/2} \sqrt{v(\hat{\theta}) / n}$ has asymptotic coverage $1 - \alpha$. For the MLE, $v = I^{-1}$. For a proportion, $\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}(1-\hat{p}) / n}$.

Proof. $\sqrt{n}(\hat{\theta} - \theta) / \sqrt{v(\hat{\theta})} = [\sqrt{n}(\hat{\theta} - \theta) / \sqrt{v(\theta)}] \cdot [\sqrt{v(\theta) / v(\hat{\theta})}]$; the first factor tends in distribution to $\mathcal{N}(0, 1)$ and the second in probability to 1 by consistency and continuity. Apply Slutsky. $\square$

Remark. The Wald interval for a proportion has coverage that oscillates and can be far below $1 - \alpha$ when n is small or p is near 0 or 1. The Wilson interval, obtained by inverting the score test instead of replacing p by $\hat{p}$ in the variance, is the standard fix.

8 Hypothesis testing

In the notes: Hypothesis Tests

8.1 Framework

A test of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is a rule $\varphi(X) \in \{0, 1\}$ (1 means reject), typically $\varphi = \mathbb{1}\{T(X) \in R\}$ for a statistic T and rejection region R .

DEFINITION The size of the test is $\sup_{\theta \in \Theta_0} P_\theta(\varphi = 1)$; it has level α if the size is at most α . The power function is $\beta(\theta) = P_\theta(\varphi = 1)$, so the Type II error at $\theta \in \Theta_1$ is $1 - \beta(\theta)$.

DEFINITION The p -value of observed t_{obs} for a statistic T that rejects for large values is $p = \sup_{\theta \in \Theta_0} P_\theta(T \geq t_{\text{obs}})$.

LEMMA Uniformity of the p -value If H_0 is simple, T has a continuous distribution function G under H_0 , and rejection is for large T , then $p(T) = 1 - G(T) \sim \text{Uniform}(0, 1)$ under H_0 . Hence rejecting when $p \leq \alpha$ has size exactly α .

Proof. $G(T) \sim \text{Uniform}(0, 1)$ by the probability integral transform: $P(G(T) \leq u) = P(T \leq G^{-1}(u)) = u$. Then $1 - G(T)$ is also uniform. $\square$

8.2 The t -tests

THEOREM One-sample and Welch t -tests (i) For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d., under $H_0 : \mu = \mu_0$ the statistic $T = (\bar{X} - \mu_0)/(S/\sqrt{n})$ has the t_{n-1} distribution, so rejecting for $|T| > t_{n-1, 1-\alpha/2}$ has size α . (ii) For two independent normal samples with means μ_1, μ_2 , and $H_0 : \mu_1 = \mu_2$,

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \quad (15)$$

is approximately t_ν with

$$\nu = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1 - 1) + (S_2^2/n_2)^2/(n_2 - 1)}. \quad (16)$$

Part (i) is the pivot theorem with $\mu = \mu_0$. In (ii) the denominator is a weighted sum of independent scaled χ^2 variables, which is not exactly a scaled χ^2 ; the Welch–Satterthwaite ν matches the first two moments of a scaled χ_ν^2 to it. With equal variances the pooled test is exactly $t_{n_1+n_2-2}$, but Welch is robust to unequal variances at little cost.

THEOREM Duality of tests and intervals If, for each θ_0 , $A(\theta_0)$ is the acceptance region of a level- α test of $\theta = \theta_0$, then $C(x) = \{\theta_0 : x \in A(\theta_0)\}$ is a $(1 - \alpha)$ confidence set. Conversely, from a confidence set one obtains a test by rejecting $\theta = \theta_0$ iff $\theta_0 \notin C(x)$.

Proof. $P_{\theta_0}(\theta_0 \in C(X)) = P_{\theta_0}(X \in A(\theta_0)) \geq 1 - \alpha$, for every θ_0 . The converse is the same identity read backwards. $\square$

8.3 Power and sample size

THEOREM Power of the one-sided z -test Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with σ known, and reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu > \mu_0$ when $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma > z_{1-\alpha}$. Then

$$\beta(\mu_1) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma), \quad (17)$$

and $\beta(\mu_1) \geq 1 - \beta_0$ holds iff $n \geq ((z_{1-\alpha} + z_{1-\beta_0})\sigma/(\mu_1 - \mu_0))^2$.

Proof. Under $\mu = \mu_1$, $\sqrt{n}(\bar{X} - \mu_1)/\sigma \sim \mathcal{N}(0, 1)$, so $Z = Z_0 + \sqrt{n}(\mu_1 - \mu_0)/\sigma$ with $Z_0 \sim \mathcal{N}(0, 1)$. Hence $P(Z > z_{1-\alpha}) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma)$. Requiring this to be at least $1 - \beta_0$ means $z_{1-\alpha} - \sqrt{n}\delta/\sigma \leq -z_{1-\beta_0}$ (with $\delta = \mu_1 - \mu_0$), and solving for n gives the bound. $\square$

The required n scales like σ^2/δ^2 : halving the detectable effect quadruples the sample.

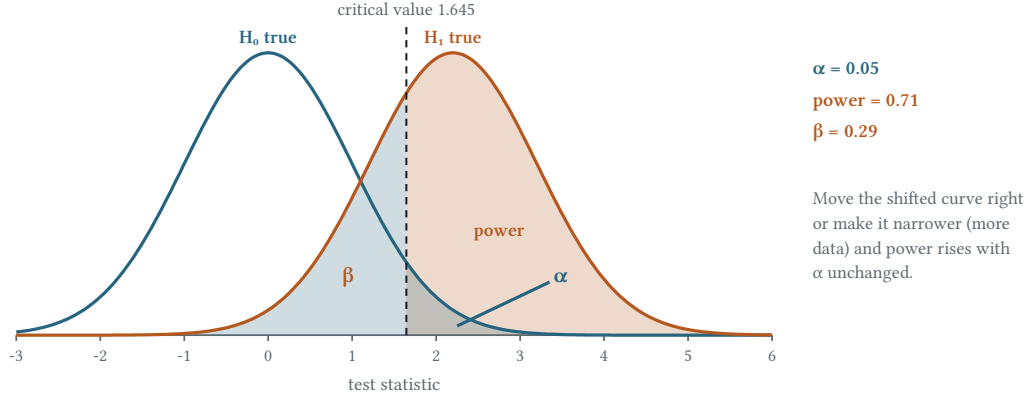


Figure 8: One-sided z -test at $\alpha = 0.05$ when the true mean is 2.2 standard errors above the null. The blue sliver beyond the line is the Type I error rate, the orange area to its right is the power, and the orange area to its left is the Type II error rate.

8.4 Likelihood ratio tests

THEOREM Neyman–Pearson lemma For simple $H_0 : \theta = \theta_0$ against simple $H_1 : \theta = \theta_1$, let φ^* reject when $L(\theta_1)/L(\theta_0) > k$, with k chosen so that φ^* has size α . Then for any other test φ of size at most α , $\beta_\varphi(\theta_1) \leq \beta_{\varphi^*}(\theta_1)$.

Proof. For every x , $(\varphi^*(x) - \varphi(x))(f(x; \theta_1) - kf(x; \theta_0)) \geq 0$: where $\varphi^* = 1$ the second factor is positive, where $\varphi^* = 0$ it is non-positive and the first factor is $-\varphi(x) \leq 0$ or zero. Integrating over x gives $\beta_{\varphi^*}(\theta_1) - \beta_\varphi(\theta_1) \geq k(\alpha_{\varphi^*} - \alpha_\varphi) \geq 0$, since both sizes are at most α and φ^* has size exactly α . $\square$

THEOREM Wilks' theorem Let $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$, where Θ has d free parameters and Θ_0 has $d - r$. Under the regularity conditions and H_0 ,

$$-2 \log \Lambda \xrightarrow{d} \chi_r^2. \quad (18)$$

The proof expands ℓ quadratically around the MLE and uses asymptotic normality of the MLE, so that $-2 \log \Lambda$ becomes a quadratic form in r asymptotically independent standard normals. The Wald and score tests are asymptotically equivalent to it. In the linear model with Gaussian errors, the likelihood ratio is a monotone function of an F statistic, which is why the F -test of regression has an exact distribution.

8.5 Multiple testing

With m tests and all nulls true, each at level α , the expected number of false rejections is $m\alpha$.

THEOREM Bonferroni and Benjamini–Hochberg (i) Rejecting each hypothesis with $p_i \leq \alpha/m$ controls the family-wise error rate, $P(\text{at least one false rejection}) \leq \alpha$, with no assumption on the dependence of the p -values. (ii) Order the p -values $p_{(1)} \leq \dots \leq p_{(m)}$ and reject the k smallest, where $k = \max\{i : p_{(i)} \leq i\alpha/m\}$. If the p -values are independent, the false discovery rate $E[V / \max(R, 1)]$ (with V false rejections out of R) is at most $\alpha m_0/m \leq \alpha$, where m_0 is the number of true nulls.

Proof. (i) By the union bound, $P(\bigcup_{i \in H_0} \{p_i \leq \alpha/m\}) \leq m_0\alpha/m \leq \alpha$. (ii) is the Benjamini–Hochberg (1995) theorem; the proof conditions on the other p -values and is omitted. $\square$

9 Summary of the main formulas

QUANTITY	RESULT	WHERE EXPLAINED
Standard error of the mean	$\sigma/\sqrt{n}$	Sampling distributions
S^2 unbiased	$E S^2 = \sigma^2$, divisor $n - 1$	Point estimation
Normal sampling laws	$(n - 1)S^2/\sigma^2 \sim \chi_{n-1}^2$, $T \sim t_{n-1}$, $\bar{X} \perp S^2$	Sampling distributions
MSE	$\text{Var} + \text{bias}^2$	Point estimation
Cramér–Rao	$\text{Var}(\hat{\theta}) \geq 1/(nI(\theta))$	Point estimation
MLE	$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$	Point estimation
CI for the mean	$\bar{X} \pm t_{n-1, 1-\alpha/2} S/\sqrt{n}$	Confidence intervals
CI for the variance	$[(n - 1)S^2/\chi_{n-1, 1-\alpha/2}^2, (n - 1)S^2/\chi_{n-1, \alpha/2}^2]$	Confidence intervals
Wald interval	$\hat{\theta} \pm z_{1-\alpha/2} \hat{\text{se}}$	Confidence intervals
Sample size, one-sided z-test	$n = ((z_{1-\alpha} + z_{1-\beta})\sigma/\delta)^2$	Hypothesis tests
Likelihood ratio	$-2 \log \Lambda \xrightarrow{d} \chi_r^2$	Hypothesis tests
Bonferroni / BH	$p_i \leq \alpha/m$ / largest k with $p_{(k)} \leq k\alpha/m$	Hypothesis tests

10 References

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